Elijah Zhao

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Education

University of British Columbia GPA: 86% | CS GPA: 91%
Bachelor of Science, Major in Computer Science, Minor in Math *Expected Graduation: May 2028*

Experience

Hootsuite May 2026 – Dec 2026
Software Engineer Intern, Platform Infrastructure *Vancouver, BC*

- Enabled clients to query Hootsuite Analytics APIs using **SQL** in Tableau, serving **50,000+ queries** through a **Java** JDBC connector using **Apache Calcite**, exposing data as tables and optimizing queries with **filter pushdown**.
- Built **OAuth/PKCE** and **dynamic client registration** in **Go** for Hootsuite's new **MCP discovery platform**.
- Developed the **first CI-gated Playwright E2E suite** for an MCP connector platform spanning three services; mocked flaky MCP and OAuth dependencies in a Docker Compose environment and covered **10+** failure paths.
- Tracked Analytics API usage with **Go** middleware, streaming request events through **Kafka** to Mixpanel.

Nexera Robotics (acquired by Locus Robotics) January 2026 – April 2026
Software Engineer Intern *Vancouver, BC*

- Led redesign of a robotic pick-cycle pipeline in **Python/C++**, parallelizing ML inference and NumPy-based decision logic with Ray Core and optimizing critical paths in **C++**, improving throughput from **7 to 10 picks/min**.
- Built internal robotics tooling in **Python/PostgreSQL** interfacing with **Universal Robots** firmware, shipping **5+** features including custom speed/pathing controls and performance tracking to identify system bottlenecks.
- Developed a dataset curation pipeline that reduced **1M+** unfiltered images to **7K** using **DINOv2** embeddings and **pgvector** similarity search for deduplication and sampling; stored images in S3 and metadata/vectors in **AWS** RDS.

UBC BizTech August 2025 – Present
Software Development Lead *Vancouver, BC*

- Leading **9** developers in building software and robotics platforms for UBC's largest tech club, serving **2,000+** users.
- Built a serverless backend with **AWS Lambda + Cognito** to manage authentication, payments, and event workflows; integrated **Stripe** webhooks to process **$30K+** in membership payments and event confirmations.
- Designed **DynamoDB** access patterns with **GSIs** enabling queries across **9+** events and **900+** members.
- Shipped custom event products, including a membership portal, NFC networking system, and mock stock exchange.

University of British Columbia January 2026 – April 2026
Teaching Assistant, Data Structures & Algorithms (CPSC 221) *Vancouver, BC*

- Mentored **110+** students during lectures and weekly labs, teaching **data structures and algorithms** in **C++**.

Projects

NBA 3D Scene Reconstruction | *Python, PyTorch, OpenCV, RF-DETR, SAM 2, Three.js* Jul 2026 - Present

- Built a **computer vision** pipeline reconstructing monocular NBA clips as interactive 3D scenes in **Three.js**.
- Combined **RF-DETR/SAM 2** detections with SigLIP jersey embeddings to track and cluster players by team.
- Used court-keypoint homography and MotionBERT 3D pose lifting to reconstruct and animate player meshes.

HumanCraft (Hack The North 2026) | *Swift, Python, Java, LiDAR, Minecraft* Sep 2026

- **Won** Best Use of **Sentry** among **1050+ hackers**. Built a Swift app capturing human point clouds from iPhone **LiDAR** and RGB data, rendering them in Minecraft. Used MediaPipe to track hands for punching interactions.

BizBot (NwHacks 2026) | *Supabase, YOLOv8, FastAPI, Next.js, Google Gemini* January 2026

- **Won** Best Real-World Ready AI Product from **Block** among **630+ hackers** by building an autonomous robot photographer with a **YOLOv8** vision loop to auto-trigger photo capture and publish an AI-filtered live gallery.

NBAsalary | *Spring Boot, JPA, Python, Java, React, TailwindCSS* August 2025 – September 2025

- Built a **Spring Boot** NBA salary API with **JPA/PostgreSQL** for relational data modeling and a React frontend.

Technical Skills

Languages: Python, C++, Go, JavaScript/TypeScript, Java, SQL (PostgreSQL + pgvector)
Frameworks & ML: FastAPI, React.js, Node.js, Ray, YOLOv8, OpenCV, RF-DETR, SAM 2, PyTorch, Spring Boot
Cloud & Tools: AWS (Lambda/DynamoDB/EC2/Amplify/Cognito/API Gateway), Kafka, Docker, Git, Linux